

Blind Audit of J-Lens / R-Lens / Logit-Lens on Qwen3.6-27B

Executive Summary (DRAFT - agent synthesis, provisional). Final exec summary & pivot are the human's.

1. Question and hypotheses

Does a lens's signal in a blind scan survive when the text-reconstruction channel is neutralized?

Three competing hypotheses are discriminated:

- H1 - internal readout: the lens surfaces a genuine internal signal, beyond surface text.
- H2 - text inversion: the apparent signal is just the scan letting a reader reconstruct the text.
- H3 - judge apophenia: the judge manufactures 'anomalies' from anomaly-shaped noise.

2. Method

11 human-validated anomalous/clean text pairs (families: bug, false_premise, injection). Each pair is scanned by 3 instruments (J-lens, R-lens, logit lens) into per-position, per-layer top-k token stacks. Two LLM judges (judgeA = claude-sonnet-4.6; judgeB = gpt-4.1-mini) score each scan for anomaly + confidence, blind to any anomaly family or instrument name. Seven fixed conditions include the controls that carry the argument: prompt-only (text baseline), blind scan, scan_permuted (positions shuffled = H3 control), and reconstruction (judge A rebuilds text from the scan, judge B reads only that = H2 control). A leak audit gates every scan run.

3. Headline results (all figures/numbers computed this session; see Result.md)

Leak gate GREEN: 0 shared 4-grams across 66 scans; no text-reconstruction leak. 579/592 valid verdicts (2.2% loss). Independent from-scratch AUC recompute coincides with the pipeline to $<1e-6$.

- Text baseline is near-ceiling: prompt-only AUC 0.95 (judgeA) / 0.96 (judgeB).
- Pooled blind-scan AUC per instrument: logit 0.84 $\geq$ rlens 0.80 $\geq$ jlens 0.75 (n=11/11).
- Triplet (dAUC, 95% paired-bootstrap CI), pooled over judges/prompts:
 - D1 scan - prompt: jlens -0.21 [-0.47,0.01] | rlens -0.16 [-0.30,-0.02] | logit -0.12 [-0.30,0.05]
 - D2 scan - reconstruction: jlens +0.11 [-0.30,0.47] | rlens +0.49 [0.23,0.79] | logit +0.32 [0.05,0.57]
 - D3 permuted - chance: jlens +0.28 [0.02,0.46] | rlens +0.20 [0.01,0.39] | logit +0.29 [0.11,0.45]
- Inter-judge agreement (scan, v1): Cohen kappa 0.49, Spearman 0.50 (moderate).
- Evidence fidelity: ~52% of tokens the judge CITES are absent from the scan; ~13% of the present ones fall in the anomaly zone. High false-alarm on CLEAN twins (reconstruction 0.5-1.0).

4. What the data asserts (factual)

- D3 > 0 with CI excluding 0 on all three instruments: shuffling positions does NOT destroy the discriminative signal -> a large part of it is order-invariant.
- D1 < 0 everywhere: the blind scan never beats reading the text.
- The logit lens - which has no privileged internal-readout machinery - gives the HIGHEST blind-scan AUC, matching/beating the two workspace lenses.
- The judge fabricates the majority of its cited evidence and flags clean twins often.

4b. Content control (judge-independent, causal) - the decisive addition

Does the anomaly leave an objective fingerprint in the scan (not just via the judge)? On the 5 length-aligned single-token pairs, compare the anomalous vs clean scan top-k per position, split by region relative to the anomaly at position s. Under causal attention only positions $\geq s$ can be affected; positions $> s$ carry the SAME input token in both twins, so any difference there is the anomaly's internal contextual propagation. Fraction of (position, layer) cells that differ:

- upstream ($< s$): 0.000 / 0.000 / 0.000 (early/mid/late) - causal sanity check PASSED (exactly 0).
- at ($= s$): 0.956 / 1.000 / 1.000 - trivial: the input token itself differs (echo). Ignore.

- downstream (>s): 0.254 / 0.602 / 0.720 - SAME input token -> internal propagation of the anomaly. So a one-token anomaly measurably changes the lens scan at downstream positions (identical input there) in 25% of early-layer and 60% of mid-layer cells - an objective, causal, judge-independent fingerprint, past the late-layer echo. It is NOT lens-specific: the logit lens propagates as much (0.20/0.64 early/mid) as J/R. Caveat: 'differ' means the top-k changed, not that it changed in an anomaly-INTERPRETABLE way - it proves the information exists and propagates, not that the judge reads it.

5. Bugs and anomalies encountered (and fixed)

- judgeB (gpt-5-mini) reasoning-truncated EVERY JSON verdict (finish_reason=length, empty) -> switched to gpt-4.1-mini (non-reasoning, still OpenAI family).
- Judge client had no timeout -> a stalled call hung the whole run; max_tokens=600 too low for claude on verbose scans -> added timeout=90 and raised to 1200.
- content_filter: claude refuses to read some reconstructed texts, biased onto false_premise (~10 lost keys) -> makes D2 fragile.
- Windows cp1252 vs UTF-8 crash on every scan read -> fixed at the root (encoding='utf-8').

6. Limitations / caveats

- n = 11 pairs: pooled CIs are wide; per-family AUCs are exploratory noise.
- The permutation control preserves token CONTENT (only order is shuffled), so D3>0 cannot separate apophenia (H3) from a real-but-position-invariant readout. This is the key design gap.
- D2 compares the scan to a BROKEN reconstructor (high false-alarm, degenerate outputs, content_filter hole) -> it cannot cleanly reject H2.
- Judge B was changed mid-project (gpt-5-mini -> gpt-4.1-mini); results differ from the frozen plan.

7. Provisional conclusion (AGENT OPINION - to be verified, not the pivot)

Two-level picture. The INSTRUMENT (scan) carries a real, internal, causal signal: the content control shows a one-token anomaly propagates to downstream early/mid layers, judge-independently (causal sanity passed). This refutes 'pure apophenia' at the INFORMATION level - anomalous and clean scans are objectively distinguishable - and shows the signal is not merely surface text (H2). BUT it is NOT lens-specific (logit propagates as much as J/R), and the JUDGE exploits it poorly: blind-scan AUC is modest and never beats reading the text, is highest for the 'dumb' logit lens, shows order-invariant behaviour (D3), fabricates most cited evidence, and false-alarms on clean twins. Honest headline: a real internal signal exists and propagates, but it is not privileged to the J/R-lens machinery and is not cleanly recovered by the judge, at n=11. Apophenia is a valid critique of the judge, not of the scan's information content.

8. Potential leads / next steps

- Content control DONE and POSITIVE (see 4b): the anomaly propagates internally. Next, test whether the downstream change is anomaly-INTERPRETABLE (not just any reshuffle) - e.g. does an anomaly-trained probe read it, or does the surfaced vocabulary shift toward the anomaly semantics.
- Foreign-scan control (judge a different clean item's scan) to bound the JUDGE-side apophenia directly, now that the scan is known to contain real signal.
- Fix or drop the reconstruction channel before trusting D2 (content_filter + degeneracy).
- Classify ALL false positives (not just the 5 in the seed-0 sample) into (a) evokes family / (b) evokes own content / (c) invents; proportion (c) is the direct H3 estimate.
- Scale n beyond 11 if budget allows; report only pooled AUC + triplet as primary.

9. Budget and artifacts

Spend 14.97/19. Artifacts:

results/{metrics,triplet,inter_judge,leak_check,evidence_check,family_false_alarm,blockH_worksheet},
figs/fig1_triplet.png, fig2_auc_family.png, judge/outputs/verdicts.jsonl, Result.md (full record),
experiments.md entry (14).

Figure 1 - H1/H2/H3 triplet per instrument (95% paired-bootstrap CI)

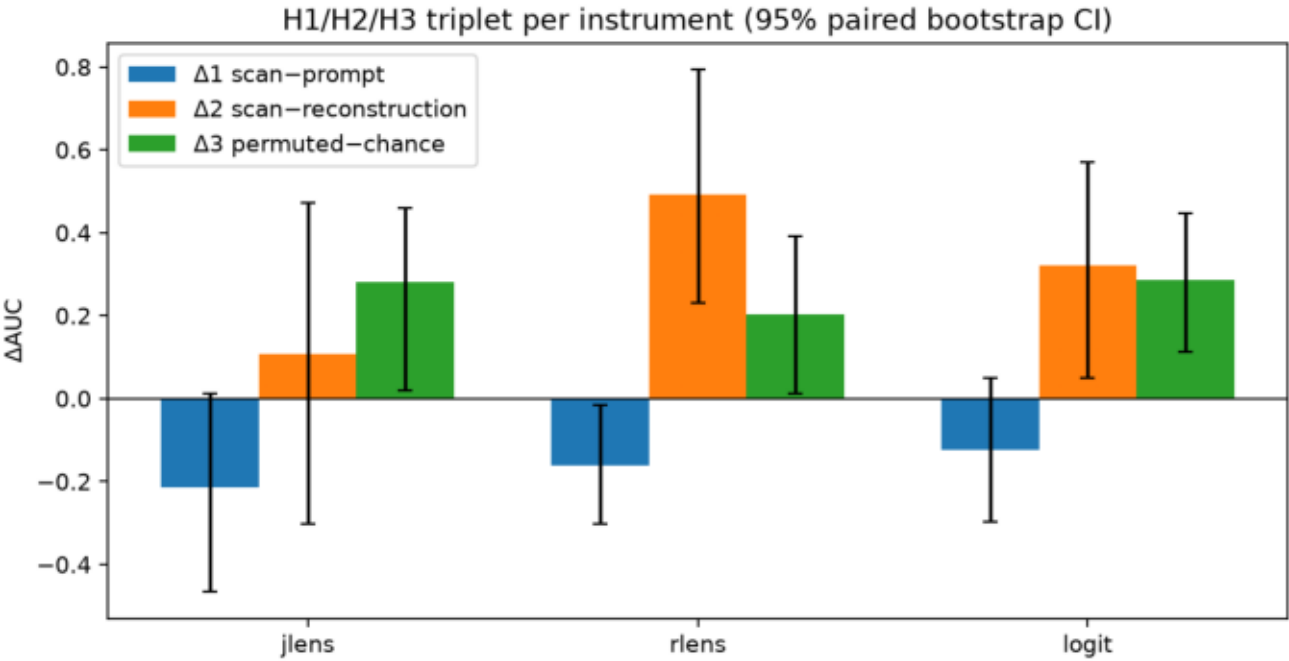


Figure 2 - AUC per family x instrument (exploratory, n~4/family)

